



FRANCISCO ARAÚJO

M.Sc. AEROSPACE ENGINEERING

• Open to relocation — Silverstone, UK

CONTACT

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LOCATION

Guimarães, Portugal

EDUCATION

M.Sc. Aerospace Engineering

University of Minho · Exp. Jul 2027

GPA 3.5 / 4.0

B.Sc. Aerospace Engineering

University of Minho · 2023–2025

B.Sc. Aeronautical Engineering

UBI — Covilhã · 2021–2023

CAD / CAE

CATIA V5	
ANSYS	
SolidWorks	
Nastran	
Solid Edge	

TECHNICAL TOOLKIT

Propulsion / CFD	
Aircraft Structures	
Aerodynamics	
MATLAB / Simulink	
Python	

Also: COMSOL, OpenFOAM, STK, KiCad

LANGUAGES

Portuguese — Native

English — Fluent (C1)

CORE STRENGTHS

- Team leadership under deadline pressure
- Cross-functional communication
- Problem-solving & design iteration
- Financial oversight & governance
- Event & project logistics

PROFESSIONAL PROFILE

Aerospace engineering Master's student combining hands-on propulsion leadership, aerodynamic CFD/ML research, and precision CAD delivery across CATIA V5, ANSYS and Nastran. As Head of Propulsion at MAST, leads full-cycle solid-motor development for Europe's premier university rocketry challenge (EuRoC); as a CEiiA intern, contributed structural design to Portugal's first nationally designed aircraft. Brings a racer's competitive drive; first sparked leading a national F1 in Schools campaign paired with the rigour and discipline of formal aerospace engineering training.

MIT PROGRAMMES

MIT Portugal Innovation Workshop

Massachusetts Institute of Technology, Cambridge, MA · Jun 2026

- ▶ Selected as one of 35 Portuguese students and researchers for MIT Portugal's venture-creation programme.
- ▶ Trained in evaluating commercial potential, target-market selection, product/market-fit de-risking and building founding teams, through seminars led by MIT and Portuguese entrepreneurs.

Manufacturing: From Fundamentals to the Future

MIT — Center for Advanced Production Technologies & Dept. of Mechanical Engineering, Cambridge, MA · Jul 2026

- ▶ One of 24 students selected worldwide (12 MIT, 12 Portugal) for an intensive two-week hands-on manufacturing course built on MIT's 2.008.
- ▶ Delivered a scaled manufacturing project spanning injection moulding, thermoforming and metrology, then a competition-based additive-manufacturing project using computational design and advanced 3D printing.
- ▶ Explored production planning, AI for design and operations, factory simulation and digital twins.

KEY ENGINEERING EXPERIENCE

Head of Propulsion Department

MAST — EuRoC University Rocketry Team, Guimarães · Nov 2024 – Present

- ▶ Own full-cycle design of a KNSB solid-propellant rocket motor for the S3 category, targeting a 3,000 m apogee at Europe's top university rocketry competition.
- ▶ Coordinate a multidisciplinary propulsion sub-team, aligning design trade-offs with structures, avionics and recovery leads under competition deadlines.

Structural Design Intern — “Shaping the Future Aeronautics”

CEiiA, Évora · Jul – Aug 2024

- ▶ Contributed to cockpit structural development for the LUS 222, Portugal's first nationally designed aircraft, inside a live OEM engineering workflow, using CATIA V5 and Nastran.

Vibration Attenuation Trainee

IEEE Student Chapter, Guimarães · Oct 2024 – Present

- ▶ Researching passive and active vibration attenuation strategies for UAV powertrains, using modal analysis and FRF testing to characterise dynamic behaviour.

ACADEMIC RESEARCH

M.Sc. Dissertation — Thrust-Vector Control for Trim-Drag Reduction

University of Minho · Expected Jul 2027

- ▶ Developing an AI-assisted aerodynamic optimisation framework (CFD + ML surrogate models) applied to a pusher-configuration UAS to reduce reliance on trim control surfaces.

LEADERSHIP & EXTRACURRICULAR

Director for Sustainability — AAUMinho, Guimarães Jan 2025 – Jan 2026

| Directs environmental and social-responsibility projects across the student body; builds NGO and academic partnerships.

President of the Fiscal Council — AeroUM, Guimarães Oct 2025 – Present

| Independent financial oversight and auditing, ensuring institutional compliance with university and legal regulations.

Vice-Director for Science & Recreation — AeroUM, Guimarães Oct 2024 – Oct 2025

| Designed the association's scientific and recreational programme, organising workshops and talks with industry professionals.

Team Leader — F1 in Schools — Felgueiras Dec 2020 – Jun 2021

| Led design and manufacture of a miniature F1 car to national-stage qualification.